

USB 2.0 to Isolated RS-485 / RS-422 Converter

PROJECT OVERVIEW

- This project implements a robust USB 2.0 to RS-485 / RS-422 communication interface intended for industrial and embedded applications.
- The design provides two independently isolated communication channels for RS-485 and RS-422 operation. Interface selection does not require mechanical switches or jumpers. The receiver paths are automatically combined by utilizing the guaranteed fail-safe HIGH state of the RS-485/RS-422 transceivers.
- Galvanic isolation is provided for both communication signals and bus-side power. The RS-485 interface includes software-controlled 120 Ω termination

KEY FEATURES

- USB 2.0 Full-Speed interface
- FT230X USB-to-UART bridge supporting up to 3 Mbaud
- Independent isolated RS-485 and RS-422 channels
- ADM3065E true fail-safe RS-485/RS-422 transceivers
- Automatic receive-path selection without mechanical switches
- Integrated signal and power isolation
- Software-controlled 120 Ω termination via RTS
- Common-mode filtering on communication lines
- Configurable external fail-safe biasing
- Multi-stage RS-485 protection using SM712 + TBU
- Optional thyristor surge protection stage
- Protection and filtering components configurable for EMC optimization
- USB-powered operation

		USB to RS485_RS422 Converter	
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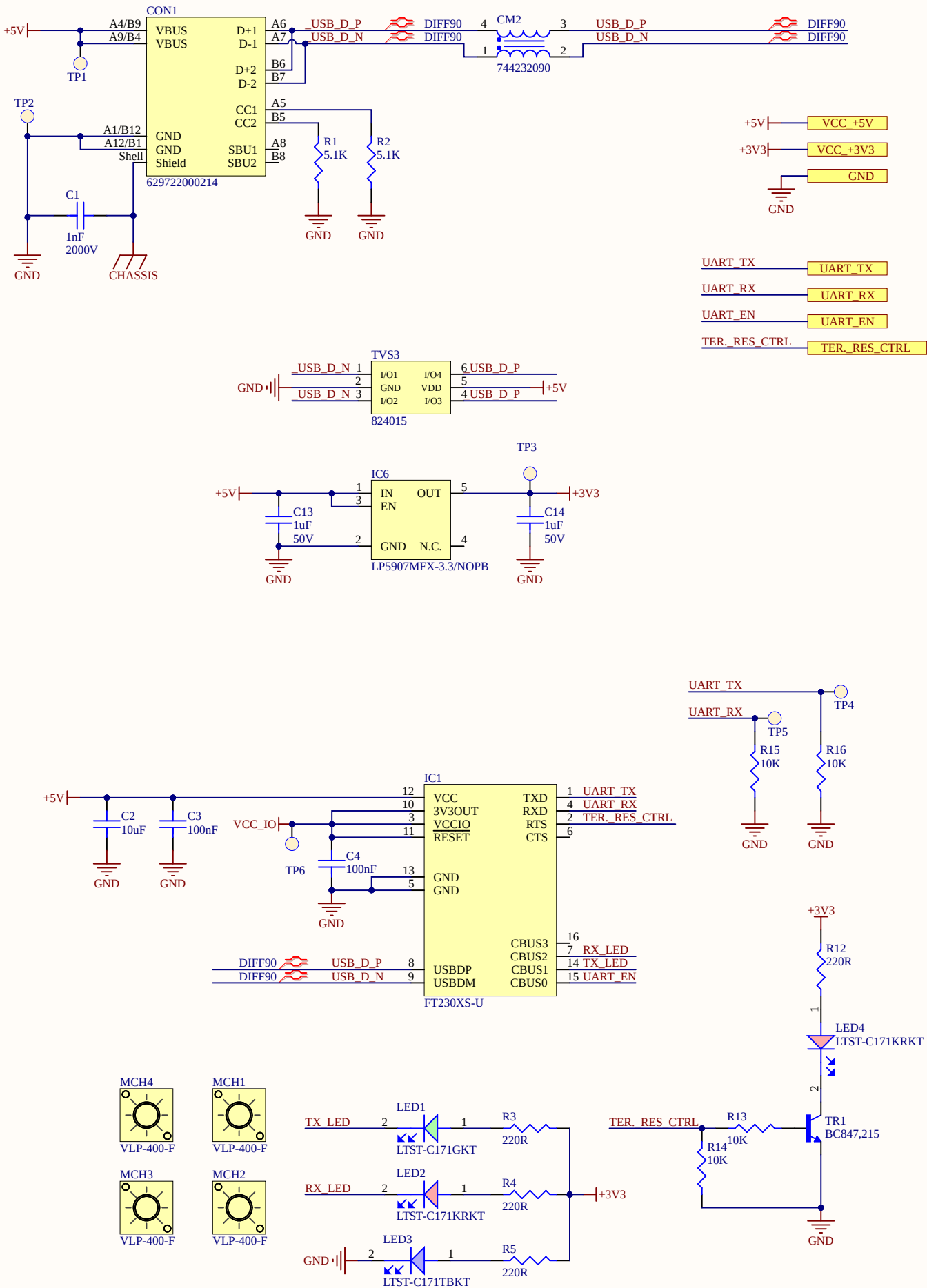
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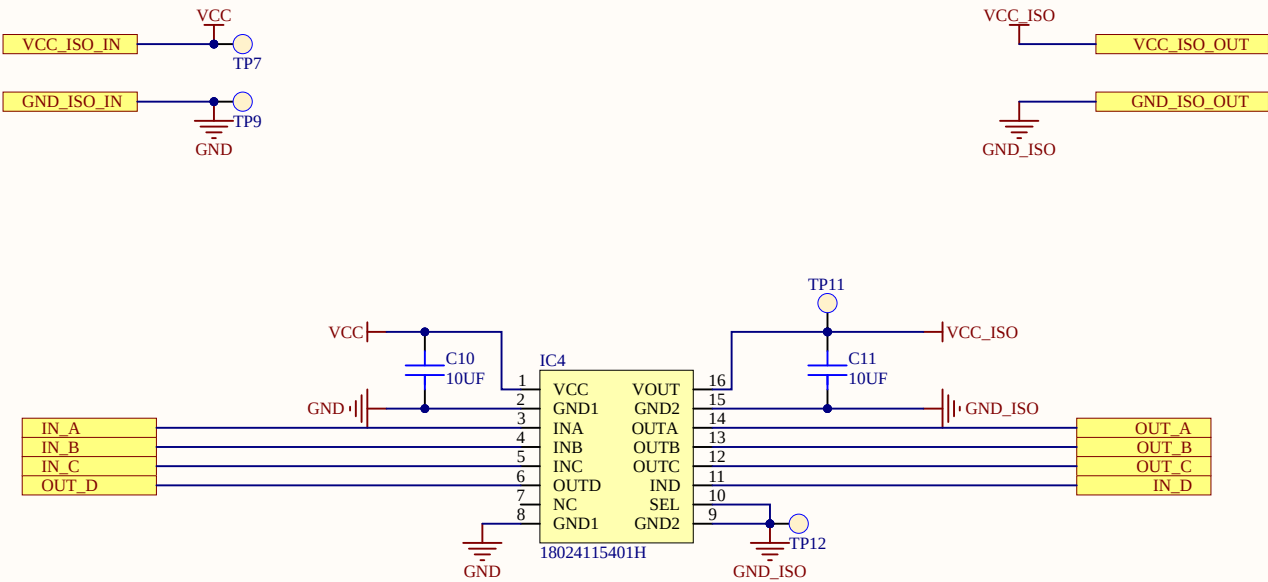
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